

The cultural evolution of human goals: How individuals generate, select, and transmit goals

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Abstract

Humans pursue goals that are remarkably diverse and vary over time and cultures. These goals shape which behaviors are explored, valued, and socially transmitted, yet most theories of cultural evolution focus on how behaviors evolve while leaving the origins of goals unexamined. We argue that a complete understanding of cultural evolution requires explaining how goals themselves emerge, vary, and persist across generations. Building on studies of motivation and curiosity in cognitive science and artificial intelligence, we introduce the notion of cultural autotelic agents—individuals who actively generate, select, and transmit their own goals within social environments. By highlighting the cognitive and motivational mechanisms that drive goal formation and selection, this framework extends existing models of cultural evolution and helps explain the open-ended, self-propelling character of human culture.

1 Introduction

Humans produce an extraordinary diversity of cultural artifacts and practices—from cave paintings to the Olympics games or space travel—that accumulate and transform across generations. Cultural evolution theory seeks to explain how such diversity arises, focusing on how behaviors, knowledge, and technologies vary, spread, and improve over time (Boyd and Richerson, 1985; Henrich, 2015; Mesoudi and Thornton, 2018). Yet behind most human feats lies a *goal*: a representation of an imagined future state that individuals are motivated to achieve (Elliot and Fryer, 2008). For instance, the goal of “getting a man on the moon” is what drove collective innovation in the specific direction that resulted in the Apollo 11 mission. In this paper, we turn the lens toward the cultural evolution of goals. Where do human goals come from? How are they created, shared, and selected across generations? And how can their diversity—and occasional eccentricity—be explained?

Across history and cultures, human goals have followed rich and diverse dynamics. Some, like predicting the weather or building flying machines, have persisted for centuries (Chou, 1979; Velazquez, 2016). Others, such as the dream of transmuting metals into gold, inspired generations before disappearing entirely (Principe, 2011). And some, like perfecting spinning tops, have endured for millennia despite little practical use (Piacentini and Delli Castelli, 2023; Best, 1976). New goals continue to emerge today in similarly unexpected forms: from Matsés children in the Amazon devoting time to learn fishing while their parents’ activities are forest-oriented (Lew-Levy and Amir, 2024), to runners creating drawings on digital GPS maps through elaborated routes (Cantor, 2019).

Historically, however, the concept of a goal has been largely absent from discussions in cultural evolution research. Both empirical and theoretical studies have rather focused on how behaviors spread and improve across generations, leading, for instance, to a better understanding of the mechanisms supporting their cultural transmission (Lewis and Laland, 2012; Caldwell and Millen, 2009; Lucas et al., 2020) and the learning strategies guiding their selection (Kendal et al., 2018; Rendell et al., 2010; Thompson et al., 2022). That goals are useful and necessary concepts to explain how behaviors culturally evolve was only made explicit in a recent paper by Singh (2020). There, Singh argues that goals shape behavioral evolution in two ways: first, by influencing which behaviors are considered valuable and targeted by selective social learning and, second, by directing exploration and innovation toward new and more efficient ways of

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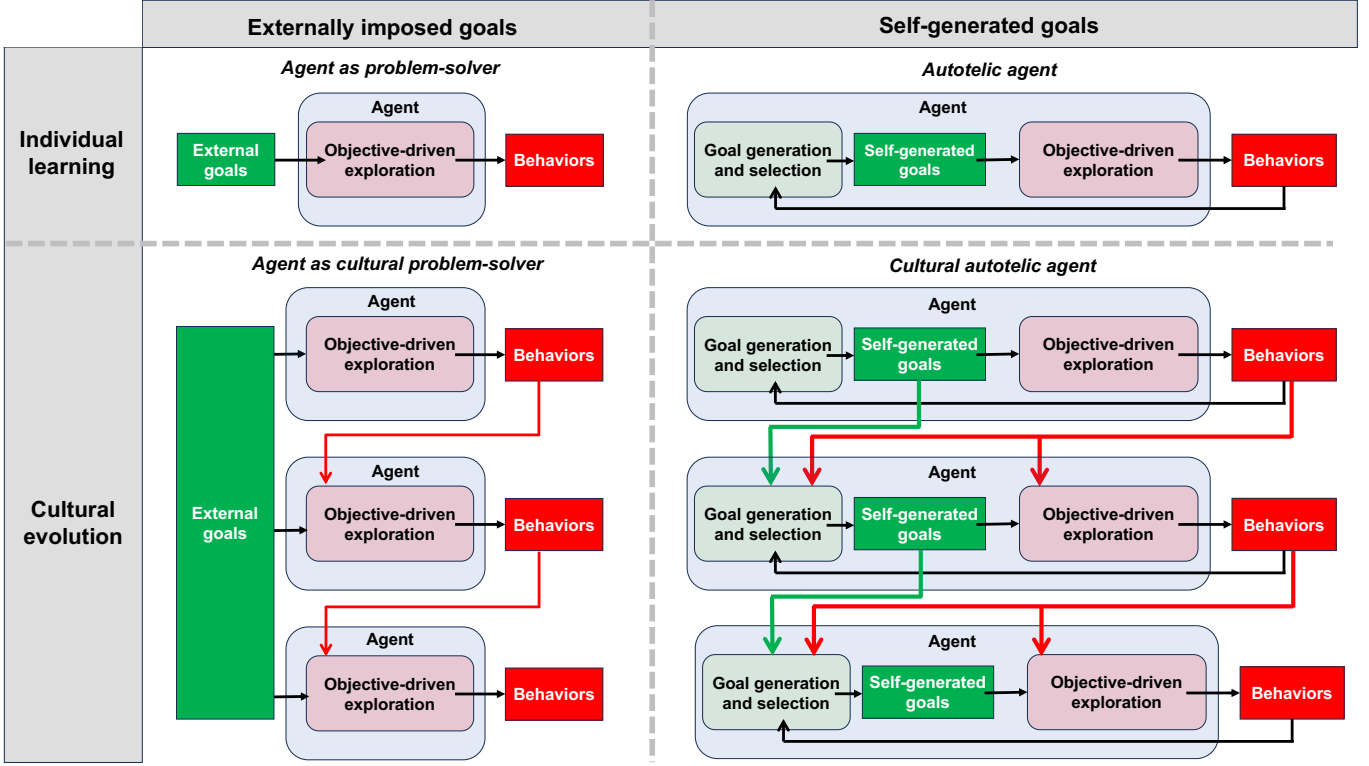


Figure 1: **Humans are cultural autotelic agents.** We introduce the notion of cultural autotelic agents, i.e. agents that combine individual and social learning to represent, generate, select, and transmit their own goals. This model departs from the historical conceptualization as problem-solvers (left column). This standard perspective focuses on how agents optimize behaviors toward goals that are externally imposed. This view is largely present in research on individual cognition (top-left) and has inspired most experimental paradigms in cultural evolution (bottom left, e.g. transmission chains). Research on motivation, developmental psychology, and developmental artificial intelligence has extended this view toward the concept of *autotelic agents*, i.e. agents able to self-generate and pursue their own goals (top-right). This conceptualization affords a more complete understanding of proactively exploratory behaviors in humans, in particular how their past behavior influence their goal generation and selection mechanisms. Here, we propose integrating such insights from cultural evolution and autotelic learning to introduce the concept of *cultural autotelic agents* (bottom-right). Under this view, agents are active in the generation and selection of the goals they pursue. These goal generation and selection mechanisms are influenced both by social information and individually collected information. We argue that this conceptualization is necessary to think about the cultural evolutionary dynamics of goals.

fulfilling desired outcomes. Under this view, goals are major causal drivers of behavioral evolution, which means we must attend to how goals themselves evolve and guide human behavior, not only to the behaviors they produce (Charbonneau et al., 2023).

To our knowledge, only two recent studies attempted to model the cultural evolution of goals (Winters, 2019; Winters and Charbonneau, 2025). Both studies model goal variation and selection mechanisms at a high level of abstraction, and explore the resulting co-evolutionary dynamics between goals and solutions. This approach is useful for exploring a broad range of scenarios, but comes at the cost of limited grounding in actual human behavior. To complement this approach, we propose deriving the mechanisms underlying goal variation, selection, and transmission from what is known about human cognition and motivation. In particular, we build on converging research agendas within cognitive science which led to the characterization of humans as autotelic individuals (from the Greek *auto* (self) and *telos* (goals)): individuals who generate and select goals according to their motivational mechanisms (Colas et al., 2022b; Sigaud et al., 2022; Molinaro and Collins, 2023; Chu et al., 2024). This perspective moves beyond the standard view of individuals as mere problem solvers to emphasize their ability to *generate problems of their own*. Appreciating the autotelic nature of individuals is crucial for grounding the study of goal evolution in individual cognition, which is the focus of Section 2. Section 3 discusses the implications of this perspective toward understanding the open-endedness of human culture. Section 4 concludes by discussing methodological approaches for empirically studying the cultural evolution of human goals.

2 Grounding a cultural evolutionary theory of goals in individual cognition

Understanding how goals evolve requires recognizing that individuals’ cognitive and motivational systems allow them to generate, select, and transmit goals. To begin outlining such a perspective, we draw inspiration from research on motivation and curiosity in developmental psychology, neuroscience, computational cognitive science, and artificial intelligence. These fields are converging on formal models of autotelic agents—systems motivated to represent, generate, transmit, and select their own goals (O’Reilly, 2020; Colas et al., 2022b; Sigaud et al., 2022; Molinaro and Collins, 2023; Chu et al., 2024). Borrowing these computational and conceptual tools allows us to describe how goal variation, selection, and transmission arise from the interaction between human motivation and social learning, providing a foundation for modeling the cultural evolution of goals.

2.1 Defining goals, behaviors, and goal-directed behaviors

In psychology, a goal is commonly defined as a “cognitive representation of a future object that the organism is committed to approach or avoid” (Elliot and Fryer, 2008, p. 244). A goal also entails an *evaluative function* specifying what counts as progress or achievement relative to that imagined future (Colas et al., 2022b; O’Reilly, 2020). Defined this way, goals are not restricted to being specific target states of the world (e.g. going to the supermarket), but can capture any kind of self-generated problem; any set of constraints on future world states (e.g. building a canoe to carry 10 people across the river by the end of the week), or future mental states (e.g. learning about XIV century Japanese history), including abstract objectives (e.g. building a thriving community) (Colas et al., 2022b; Ady et al., 2022). By contrast, *behaviors* are the observable consequences of actions. *Goal-directed behaviors* are those behaviors that are organized around achieving a specific goal. Recognizing these distinctions is essential for understanding how goals, like behaviors, can vary and evolve through cultural evolutionary processes.

2.2 A converging shift toward autotelic agents

Early conceptualizations of human behavior in psychology regarded humans as optimizers, able to refine their behaviors to better match externally specified goals (Skinner, 1965; Thorndike, 1911; Kabiraj et al., 1944; Simon, 1955). For instance, Newell and Simon used chess as a case study in their model of human problem-solving and justified this choice by the fact that “there is a well-defined goal” and that “the competitive aspects [produce] properly motivated subjects” (Newell, 1972, p. 664). Their theory thus relies on the assumption that humans have an accurate representation of a desired end state and are motivated to reach it. Similarly, in artificial intelligence, reinforcement learning practitioners train agents to achieve predefined objectives by hand-crafting a reward function (i.e. an evaluative function), then using machine learning algorithms to select actions that maximize expected future rewards (Sutton et al., 1999); in other words, the goal is simply to maximize a signal established a priori. In most experimental psychology studies, subjects are given pre-defined goals specified by the experimenter (Figure 1, top-left), whereas goals themselves are rarely the object of study (Karayanni and Nelken, 2022; Molinaro and Collins, 2023). This also applies to studies of cultural evolution, and is perhaps most salient in transmission chain experiments, which involve “participants learning from the previous participant in their chain along a series of cultural ‘generations’” (Mesoudi, 2021, p. 6). In such studies, participants might be asked to maximize the flying distance of paper airplanes (Caldwell and Millen, 2008) or the rolling speed of a wheel (Dere et al., 2019) and are rewarded based on their performance. Participants play no role in the definition and selection of the goal, and the experimenter takes care of aligning subjects’ motivation with the assigned goal (Figure 1, bottom-left).

While this conceptual choice to take goals as starting assumptions has facilitated the explanation and modeling of problem-solving, it has left the origin of the goals that drive such behaviors unexplained. Inspired by early research on intrinsic motivations and self-determination (Berlyne, 1966; Deci and Ryan, 1985), as well as theories of active learning (Piaget and Cook, 1952; Gopnik et al., 1999), recent works in cognitive science have started to explore the mechanisms underlying autonomous goal generation and selection (Chu, 2023; Molinaro et al., 2024; Chu and Schulz, 2022; Ten et al., 2021b; Poli et al., 2022). Recent developments in artificial intelligence mirror this conceptual shift, with the introduction of autotelic or goal-generating reinforcement learning frameworks, where agents internally select and pursue goals following internal learning signals rather than externally-defined reward functions (Srivastava et al., 2013; Baranes and Oudeyer, 2013; Colas et al., 2022b) (Figure 1, top-right).

This joint effort, with cognitive scientists identifying key mechanisms and computer scientists revealing their implications

for open-ended learning, equips us to develop a cultural evolutionary theory of goals grounded in individual cognition. This requires combining social learning with autotelic goal exploration, an approach that several recent studies have begun to pursue. The framework of *teachable autotelic agents* considers agents who generate their own goals but can also learn goal representations from social feedback, demonstrations, or instructions (Colas et al., 2020; Akakzia et al., 2020; Sigaud et al., 2022). More recent works leverage large language models as cultural mediators, allowing agents to generate goals shaped by shared linguistic representations and norms (Colas et al., 2023; Du et al., 2023; Zhang et al., 2023). Social information—information that individuals obtain by observing or interacting with others or their products (Rendell et al., 2011)—not only accelerates skill acquisition but also expands the space of conceivable goals by transmitting abstract representations, evaluation criteria, and values (Colas et al., 2022a).

These contributions pave the way for an integrated framework bridging autotelic learning and cultural evolution. In the next section, we build on this foundation to examine the three core processes underlying goal evolution—variation, transmission, and selection—and show how they can be conceptualized using insights from cognitive science and artificial intelligence.

2.3 Cultural autotelic agents

By integrating the concept of autotelic agents into cultural evolutionary theory, we propose a view of humans as *cultural autotelic agents*: individuals capable of combining individual and social learning to represent, generate, select, and transmit their own goals (Figure 1, bottom-right). Through imagination, analogy, and recombination, individuals generate *variation* in goals (Molinaro and Bar, 2025; Davidson et al., 2025b); through motivational mechanisms, they *select* those they find meaningful or rewarding (Molinaro and Collins, 2023; Chu et al., 2024); and through ecological inheritance and social learning mechanisms such as inference, emulation, or teaching, these goals are *transmitted* over generations (Jara-Ettinger et al., 2016; Aarts et al., 2004). We now detail each of these three processes.

Goal variation. Goals can change through a range of cognitive and social mechanisms. They may be *modified*, when transmission errors or reinterpretations alter an existing goal—e.g., when a player misunderstands the rules of a game and thereby invents a new version of it. They may be *recombined*, when constraints or subgoals from distinct activities are merged, as in hybrid games that mix the rules of table tennis and soccer (teqball), or chess and boxing (chessboxing). Goals can also be *extended* or *refined* as growing skills, knowledge, or technologies enable more ambitious or more precisely defined versions of earlier aims, as is the case with Guinness World Records. These generative processes can unfold spontaneously (driven by imagination, curiosity, or play; (Chu and Schulz, 2022; Molinaro and Bar, 2025)) or be facilitated by contextual changes that reshape what can be imagined. The mastery of engine-powered flight, for instance, expanded the reachable space of goals and made space travel a conceivable aspiration (Velazquez, 2016). Likewise, new ecological and societal contexts can prompt the formulation of novel goals. For instance, greater market integration led Zinacantec Maya to switch from traditional designs where “the goal of clothing design was to demonstrate that the wearer was a member of the community” to more creative designs where “the goal was also to identify the wearer as a unique individual” (Greenfield, 2006, p. 477). Overall, goal variation is shaped existing behaviors, contexts, representations, and individual motivations as each generation’s actions and environments influence what can be thought of, attempted, and pursued.

Goal transmission. Several cognitive and social pathways support goal transmission. Perhaps the simplest one is emulation. In contrast to imitation—where the social learner copies the actions of another individual—emulation refers to “learning from the environmental results of what an agent does” (Whiten et al., 2009, p. 2418). For instance, “if a [chimpanzee] mother rolls a log and eats the insects underneath, her child will very likely follow suit. This is simply because the child learned from the mother’s act that there are insects under the log—a fact she did not know and very likely would not have discovered on her own” (Tomasello, 1999, p. 29). Emulation can thus be seen as socially facilitated goal discovery: the goal-directed actions of one individual make a desirable end state salient to another. In emulation, the learner only acquires the actor’s goal when the actor succeeds. In many situations, however, observers can go beyond this and infer the intended goal even when the attempt fails—a broader capacity often referred to as goal inference.

Starting early on in development, humans interpret others’ actions as organized around goals, inferring what others are trying to achieve even from sparse cues (Woodward, 1998; Gergely and Csibra, 2003). This enables individuals to reconstruct and adopt objectives they are never explicitly instructed to pursue. Indeed, goals can propagate implicitly through goal contagion, whereby merely observing someone pursue an aim induces the same motivation in the observer (Aarts et al., 2004). Beyond inference and contagion, language and teaching allow goals to be transmitted deliberately (Csibra and Gergely, 2009; Heyes, 2016). Humans also possess a distinctive capacity for shared intentionality: they can

form shared goals that require mutual commitment and coordination (Tomasello, 2009, 2020). Institutions, norms, and artifacts stabilize these collective goals, allowing them to persist as shared representations beyond the individuals who originated them. These representations can exist at multiple scales—from implicit personal aspirations to group norms and fully explicit societal goals (Rigoli and Lennon, 2025). Finally, ecological inheritance likely plays a major role in the inter-generational stability of goals (Baumard and André, 2025). Just like new ecological contexts can be the source of goal variation, stable environments can explain why a new generation sets similar goals to those of the previous one.

Goal selection. One major contribution the concept of autotelic learning can bring to cultural evolution is that of highlighting motivation as a relevant cognitive mechanism for understanding goal evolutionary dynamics. Indeed, while social information may affect which goals are available, goal adoption depends on whether individuals are motivated to pursue these goals. For instance, the goal learning to play the violin, and the associated evaluation function, are available to individuals who grow up in a specific cultural context (one that contains violins and experienced violin players). However, it is the alignment of that goal with motivational mechanisms that will lead individuals to actually select this goal and invest time and energy over an extended period of time to try and reach it. Thus, the active role of individuals in the selection of the goals they pursue makes motivation a central mechanism in cultural evolution (Nichols, 2004; Davis et al., 2018; Baumard and André, 2025).

Motivation is typically classified by psychologists into two main types: intrinsic and extrinsic (Ryan and Deci, 2000). Extrinsic motivation refers to an individual’s desire to pursue a goal because of the external reward (e.g., food, money, or social praise) associated with its completion. Intrinsic motivations, by contrast, are characterized by doing an activity “for its inherent satisfaction rather than for some separable consequence” (Ryan and Deci, 2000, p. 56). A classical example of intrinsically motivating activities is solving puzzles: it is not always clear which external rewards could motivate the effort of completing a puzzle, suggesting that the motivation arises from the goal pursuit itself (Chu et al., 2024). Several theories have been proposed to characterize what makes an activity intrinsically motivating (Ten and Oudeyer, 2022; Blain and Sharot, 2021). These include optimal incongruity (Dember and Earl, 1957; Hunt, 1965), intermediate novelty (Berlyne, 1966), optimal challenge (Csikszentmihalyi, 1990; Kidd and Hayden, 2015), or learning progress (Ten et al., 2021a; Poli et al., 2022, 2025; Molinaro et al., 2024).

Extrinsic motivation undoubtedly plays a major role in shaping cultural selection of goals. For instance, the multi-generational persistence of the goal of accurately predicting weather is clearly linked to its instrumental value. In other cases, however, a goal’s tie to intrinsic motivations may help explain its cultural success. For instance, Brecher (2014) suggests that “one of the things that make a [spinning] top so engaging is its counter-intuitive behavior”. In the next section, we discuss how intrinsic motivations, acting as selective pressures on goals, may have important implications for open-ended cultural evolutionary dynamics.

3 Intrinsic motivations and open-ended cultural evolution

3.1 The puzzle of open-ended cumulative cultural evolution

A striking feature of human cultural evolution is its cumulative nature—the ability to retain and refine innovations across generations, leading to steady improvements in tools, practices, and ideas (Mesoudi and Thornton, 2018). While some non-human species show limited forms of cumulative culture (Sasaki and Biro, 2017), humans are unique in achieving *open-ended* cumulative cultural evolution, continuously improving solutions without ever reaching a performance plateau (Derex, 2022; Winters and Charbonneau, 2025; Morgan and Feldman, 2024). Existing research in cultural evolution has identified exaptation as a crucial mechanism supporting these open-ended dynamics (Winters, 2019; Andriani and Cattani, 2016). Exaptation refers to situations in which a trait or behavior originally developed for one purpose is later co-opted for another. A closely related concept is recombination, where two cultural traits are combined to produce a new solution (Derex and Mesoudi, 2020; Enquist et al., 2011). Exaptation and recombination allow for the development of solutions that could not be discovered through the gradual refinement of a pre-existing solution. For instance, early computers were built using vacuum tubes, which were previously used to detect radio waves (Stanley and Lehman, 2015). Similarly, microwave ovens were invented by re-purposing a technology initially used for radar magnetron (Andriani and Cattani, 2016). By contrast, focusing on gradually improving existing ovens would never have led to the invention of microwave ovens. While historical examples abound, it appears that exaptation requires a great deal of luck: a solution developed for a given goal must produce effects that are also relevant to achieve an unrelated goal. We propose that, while serendipity surely plays a central role, intrinsically motivated goal exploration contributes to exaptation, making it not entirely dependent on chance.

3.2 Intrinsic motivations promote exaptation and recombination

Intrinsic motivations broaden the space of explored goals. First, intrinsically motivated exploration expands the range of goals that cultural groups will pursue. For instance, Rune Elmqvist’s pacemaker was first developed as “a technical curiosity,” but later found life-saving applications when combined with other technologies such as biocompatible resins (Wagner, 2023, p.122). Similarly, the initial uselessness of number theory was proudly acknowledged by its most eminent pioneers, with Leonard Dickson allegedly said “Thank God that number theory is unsullied by any application.” (Wagner, 2023, p.122). Nevertheless, discoveries from number theory were later exapted to a wide range of goals, most famously cryptographic algorithms. In both cases, intrinsically-motivated goal exploration led to discoveries with great instrumental value once recombined with others or exapted to new problems. By broadening the range of skills and ideas of a community, intrinsically motivated exploration increases the chances that some of them will later find new and useful applications (Schimmelpfennig et al., 2021).

Intrinsic motivations target promising goals. Beyond merely broadening the space of goals, intrinsic motivations tend to specifically expand it with objectives that are especially likely to generate opportunities for exaptation. Indeed, theories of intrinsic motivations predict that goals pursued out of intrinsic motivations should be those that produce surprising effects, where new information can be gained (Ten and Oudeyer, 2022), or that are promising stepping stones for future innovations (Brändle et al., 2023). Additionally, theories such as the learning progress hypothesis (Kaplan and Oudeyer, 2004) suggest that intrinsic motivations preferentially target goals where progress is possible—that is, where downstream effects on the outside world can eventually be predicted and controlled. Therefore, beyond merely increasing goal diversity, intrinsic motivations are likely to lead to the discovery and mastery of outputs and behaviors that are especially promising sources of exaptation. Historical examples such as the interest in static electricity in ancient Greece (Derex, 2022) or the existence of wheeled toys (but not wheeled vehicles) in pre-Columbian Mexico (Ekholm, 1946) confirm that human notions of interestingness allow to discover and transmit knowledge and technologies with substantial potential for exaptation. This may be key to explaining cultural transitions that could not have emerged through reward-driven imitation alone. For instance, among past hunter-gatherer populations, spear-throwers would not have been easily supplanted by early bow prototypes, since the earliest bows were likely less efficient and would have been abandoned if efficiency had been the only criterion Derex (2022). However, curiosity about flexible materials and their ability to store elastic potential energy and release it as kinetic energy—surprising yet controllable environmental effects—may have sustained experimentation long enough for bows to eventually surpass the performance of existing spear-throwers.

While this argument about the role of intrinsic motivations for exaptation is new to the cultural evolution field, research on open-ended skill acquisition has an established tradition around this view. In computational studies, relying on intrinsic motivations has been shown to overcome exploration bottlenecks. For example, in deceptive mazes, agents rewarded for novelty rather than proximity to the exit eventually discover the correct path, while purely reward-driven agents remain stuck in dead ends (Lehman and Stanley, 2011). Similarly, curiosity-driven exploration enables children and artificial agents to keep searching and learning even when external rewards are rare or misleading (Pathak et al., 2017; Burda et al., 2018; Achiam and Sastry, 2017). While the term “exaptation” is rarely employed in this field, the concept of *transfer learning* is central and has a close meaning and refers to situations where knowledge acquired for one task accelerates learning on related tasks. Even closer to exaptation, *hindsight learning* is an explicit form of transfer learning where agents actively evaluate the outcome of a goal-directed behavior with respect to a different goal than the original one (Andrychowicz et al., 2017). Computational experiments have confirmed that combining intrinsic motivations with hindsight learning is key to enabling agents to learn complex behaviors (Andrychowicz et al., 2017; Colas et al., 2019; Akakzia et al., 2020). Overall, existing research on open-ended skill acquisition confirms the potential of intrinsically motivated goal exploration for increasing opportunities for exaptation and recombination, ultimately enabling continuous, open-ended cultural evolution.

Intrinsic motivations maintain inter-individual diversity. Beyond replicating at the cultural group level the effects documented at the individual level, the cultural selection of intrinsically motivating goals introduces an additional mechanism that is specific to collective dynamics, and that creates additional opportunities for exaptation. Namely, the fact that intrinsic motivation yields inter-individual variation in goals. Indeed, if humans were only selecting goals for their instrumental value, there should be little variation in the goals they set out to achieve: what is extrinsically motivating for one individual will often be equally motivating for another, and both should converge on the most efficient way to obtain the desired external reward. In contrast, intrinsic motivation is highly path-dependent: it depends on the evolution of skills and knowledge of an individual through time. For instance, a kid playing with cubes might be motivated in building a tower as high as possible because she has recently experienced progress toward this goal, while

another kid playing with the exact same cubes might prefer to build a house. As a consequence, intrinsic motivations may result in rather chaotic dynamics of exploration trajectories at the individual level, explaining their large diversity at the group level. Maintaining such inter-individual diversity is crucial for enabling exaptation and recombination (Smaldino et al., 2024), and previous studies have explored how specific social network structures can contribute to this diversity (Nisioti et al., 2022; Derex and Boyd, 2016; Cantor et al., 2021). By fostering path-dependent goal selection, intrinsic motivations may allow to sustain inter-individual behavioral diversity irrespectively of the social network structure.

4 Developing methods for studying culturally evolving goals

The perspective put forward in this paper highlights a number of open questions that would require empirical investigation. For example, little is known about how social information influences goal generation, or about how the structure of social networks shapes the evolutionary dynamics of goals. Current empirical approaches to cultural evolution are not directly suited to explore such questions. For instance, participants are virtually always passive recipients of goals defined by the experimenters. A very fruitful research direction may thus be to combine methodological tools used in the cognitive sciences to study autotelic learning with those traditionally used in cultural evolution experiments. We first discuss how this could allow us to study goal evolutionary dynamics through experiments, before turning to promising datasets for exploring those questions. We conclude with a discussion of the analytical tools that will be necessary to quantitatively describe and compare goal dynamics.

4.1 Goal evolution in the lab

The free-choice paradigm. A well-established experimental design to study motivation is the free-choice paradigm (Deci, 1971, 1972; Izuma and Murayama, 2013; Ten et al., 2021a; Molinaro et al., 2024; Poli et al., 2022), in which participants are provided with several learning activities among which they can freely choose. For instance, Ten et al. (2021a) used this paradigm to compare how intrinsic versus extrinsic motivation modulates goal choices. Participants were provided with four learning activities of different difficulties that involved learning an association between two stimuli. Half of the participants were told that their performance on each of the four activities would be evaluated at the end of the experiment, and that they would be rewarded accordingly (extrinsically motivated group). The other half were not told about any evaluation, and were thus free to explore as they chose (intrinsically motivated group). This experiment revealed that extrinsically motivated individuals spend more time on unlearnable activities compared to intrinsically motivated individuals. Combining the free-choice paradigm with experimental designs used in cultural evolution would allow us to explore the interaction between social learning and goal selection. For instance, combining this paradigm with the transmission chain design would allow for the study how goal selection changes over cultural generations. Combining it with closed-group designs (where participants exchange information with one another within a group (Mesoudi, 2021)) could allow to study how the interaction of extrinsically and intrinsically motivated individuals affects the collective exploration of the goal space.

Open-ended game design. While the free-choice paradigm is ideal for studying how goal selection and social learning interact, participants can only select among the set of goals or activities defined by the experimenters. Thus, it does not focus on how new goals emerge. Experimental paradigms involving open-ended game design promise to overcome this issue. This class of experiments lets participants design playful goals in open-ended environments such as Minecraft (e.g. "Jump from a mountain into a body of water") (Todd et al., 2025), a physics-based throwing domain (e.g. "Throw the ball so it hits the wall then bounces back to you") (Davidson et al., 2025a), or in the space of board games (e.g. "4 in a row wins, but only diagonal.") (Collins et al., 2025). This revealed how participants' experience with the environment affects the kind of goals they invent (Todd et al., 2025), or how they can take inspiration from examples provided by the experimenters to generate new games (Collins et al., 2025). Combining those ideas with those coming from cultural evolution experiments would allow to study how social information influences goal generation.

Picbreeder. The closest instance of an experiment combining cultural evolution with autotelic learning is perhaps *Picbreeder* (Secretan et al., 2008). In the original Picbreeder study, the authors created a website on which users could "breed" images by using simple commands whose effects were initially unknown. Users could select "parent" images as starting points to create modified "children" images, which could then be selected as parent images. Additionally, users could "publish" images they liked, making them available for others to use as their own starting points and essentially

performing a sort of selection over evolving images. From initially amorphous images, this collective exploration eventually generated many interesting and rather complex images resembling real objects and scenes such as a butterfly, a sunset landscape, an alien face, or a sports car. Picbreeder thus contrasts with classical cultural evolution experiments in a key way: there was no clear criterion dictating which image should be selected as a parent, meaning that participants were not provided with any external goal. Interestingly, the authors found that trying to re-discover an image generated by these collective dynamics with an objective-driven algorithm did not always work. Objective-driven optimization would sometimes remain stuck in a local optimum that was still pretty far from the target image. We believe such large-scale experiments of autotelic collective exploration of open-ended spaces hold great potential for studying the cultural co-evolution of goals and solutions.

4.2 Promising datasets

Beyond controlled experiments, studying real-world dataset — such as patent citations (Fleming, 2001) — has been very informative for cultural evolution. However, this was often done with a focus on behavioral outputs (e.g. how components are recombined) rather than on the goals pursued. Several emerging digital platforms now offer promising opportunities to study goal dynamics directly. Examples include GitHub¹, where software developers iteratively define and refine project objectives; Scratch², where users create and share programs around self-defined creative challenges; or the online game 1Hour1Life³, which simulates inter-generational cultural accumulation. In the latter, Vélez et al. (2024) reports a player saying *“I spent my [character’s] life entirely dedicated to taking a picture. We [the online community] already had a camera, so a huge chunk of the work was already done.”*⁴. Such an account hints at the existence of rich, longitudinal data on how accumulated cultural repertoires influence the cultural evolution of goals, allowing researchers to track how individuals generate, select, and transmit goals over time.

4.3 Analytical tools

Analyzing the evolution of goals poses unique challenges beyond those faced when studying the evolution of behaviors. The first challenge concerns *how to record goals*. Unlike behaviors, goals are internal representations that cannot be directly observed. Researchers must therefore rely on explicit descriptions to identify the goals participants pursue, or infer those from behavior. In experimental settings, participants can be prompted to describe their aims verbally, specify evaluation criteria, or provide demonstrations that can be translated into formal representations through methods such as program synthesis (Davidson et al., 2025b,a). In existing datasets, where only behavioral traces are available, goals must be inferred post hoc — an approach that introduces substantial uncertainty about the underlying goals. These difficulties also raise the question of how to represent goals computationally, from hand-coded goal spaces (Akakzia et al., 2020; Team et al., 2023), or neural networks (Colas et al., 2020), to reward-producing programs (Davidson et al., 2025b). A second challenge lies in *characterizing the evolution of goal populations through time*. Once goals can be represented, researchers need metrics that capture their diversity, complexity, and relatedness. Programmatic representations offer a promising direction, as they make it possible to project goals onto structured spaces where similarity and novelty can be quantified (Davidson et al., 2025b). Beyond static descriptions, evolutionary analyses will require methods to track goal lineages and measure their transmission, recombination, and persistence — borrowing tools from phylogenetics and population genetics.

5 Conclusion

We made the case that goals display rich and non-trivial cultural evolutionary dynamics, which existing studies have largely overlooked. As goals have a direct influence on how behaviors change over time, addressing this gap is a crucial next step for research in cultural evolution. We argued that current conceptualizations of cultural agents as mere problem-solvers are unfit to reach this explanatory target. Building on research in cognitive science and artificial intelligence, we show how appreciating individuals’ ability to autonomously set and pursue their goals is a crucial requirement for investigating goal dynamics. While here we focused on the links between intrinsically motivated goal exploration and open-ended cultural evolution, this perspective opens several other avenues for future research.

¹<https://github.com>

²<https://scratch.mit.edu>

³<https://1hour1life.net>

⁴reported in a talk at IPAM, UCLA: <https://www.ipam.ucla.edu/abstract/?tid=20600&pcode=MOIWS1>

Beyond our focus on the relevance of autotelic learning for goal cultural evolutionary dynamics, we hope that this perspective will encourage a dialogue between the fields of cultural evolution, cognitive science, and artificial intelligence (Enquist et al., 2024). Among important questions that existing works have begun to explore are the role of social curiosity in the development of social learning and in language acquisition (Oudeyer and Smith, 2016), as well as the role of culture in shaping the cognitive tools and the exploration strategies underlying curiosity-driven learning (Colas et al., 2022a). More generally, studying the reciprocal interactions between autotelic learning and cultural evolutionary dynamics (Moulin-Frier, 2022; Steels, 2011) is a crucial future direction towards a more complete account of the mechanisms underlying open-ended skill acquisition at the individual and collective levels.

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